MIL-R-25544(USAF)

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MILITARY SPECIFICATION

RECEIVING SET, RADIO AN/ARN-32
BENCH, PREFLIGHT, AND FLIGHT TESTS OF

1. SCOPE

1.1 Scope.- This specification covers methods and procedures for accomplishing the necessary performance and operational tests on Receiving Set, Radio AN/ARN-32 to determine that the equipment, both before and after installation in an aircraft, is in proper operating condition.

1.2 Articles for Test.- The articles, the testing of which is covered by this specification, are listed below. Wherever the type specification for a particular aircraft provides information not in agreement with this list, the list in the aircraft type specification shall govern. Any requirements herein which are applicable only to articles of equipment not covered in the list in the aircraft type specification shall be disregarded.

Receiver Radio R-666 ( )/ARN-32
Antenna AT-536/ARN
Mounting MT-1546/ARN-32

2. APPLICABLE DOCUMENTS

Not applicable

(Copies of documents required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

3. BENCH TESTS

3.1 General.- A permanently located test rack for mounting the marker beacon receiving set shall be provided. The rack shall be completely cabled simulating a normal installation in an aircraft and shall have a source of power similar to that supplied by the aircraft. A source of 28 volts direct current capable of furnishing approximately 1.0 ampere shall be used in subsequent bench tests specified herein.

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3.2 Test Equipment Required.- The following test equipment shall be used in performing the tests specified herein:

Multimeter AN/PSM-6, or equal
Test Oscillator BC-376-( )
Generator, Signal, AN/USM-44, or Hewlett Packard Model 608 (A 50-ohm output impedance is necessary to match the impedance of the receiver.)
General Radio Output Meter, Model 583-A, or equal
Headset HS-33-A (or HS-33), used with BC-376 ( )
Lamp Assembly AN3157-4 (amber), or equal (Any 28-volt lamp which draws approximately 150 milliamperes of current will be suitable.)
Radio Frequency Cable RG-58/U

3.3 Test Procedures.- The following test procedures shall be followed for the bench tests:

3.3.1 Visual Inspection.- The receiver, antenna assembly, and component parts shall be visually inspected for damage due to shipping or handling.

3.3.2 Warmup and Adjustment.- Install the receiver in the test rack mentioned above and turn on the equipment for at least a 5-minute warmup period. The signal generator (AN/USM-44) shall have been in operation for at least 1 hour before any tests are made on the receiver. Adjust the direct-current voltage supply to 27.5 volts at the receiver terminals. Using Test Oscillator BC-376-( ) as a heterodyne frequency meter, tune the signal generator to a frequency of 75 megacycles. Remove access cover plate on the receiver. Connect output meter, General Radio No. 583-A, or equal, to the phone jack of the marker beacon receiver using a plug PJ-055, or equal, and a two-conductor cord. Adjust output meter for a 150-ohm load.

3.3.3 Receiver Sensitivity and Output Check.- Adjust the signal generator, (AN/USM-44) for 30 percent modulation at 400 cycles per second, with the radio-frequency attenuator at minumum. Advance the radio-frequency attenuator; the indicator lamp should light at 1500 microvolts, plus or minus 10 percent. Ground pin No. 1 on the receiver; the indicator lamp should light at 500 microvolts plus or minus 10 percent. With the indicator light operating at either 1500 or 500 microvolts radio-frequency input, the power output meter should read 1 milliwatt, plus or minus 10 percent. Repeat these steps with the modulation adjusted for 1300 cycles per second and again at 3000 cycles per second. Failure of the

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indicator lamp to operate indicates a malfunction of the receiver or the test equipment used. Adjustment of the radio frequency sensitivity controls and the audio output controls in the receiver are not permitted under any circumstances. Need for such changes is evidence of defective material and should be reported as such.

3.3.4 Antenna Test.- Antenna AT-536/ARN shall be checked prior to installation in the aircraft in order to ascertain that the assembly is properly tuned to the marker beacon frequency of 75 megacycles. Place the antenna assembly in an aluminum ground screen having minimum dimensions of 4 feet by 4 feet. Connect the antenna to the marker beacon receiver using a short length of Radio Frequency Cable RG-58/U. Place Test Oscillator BC-376-( ) in operation near the antenna assembly with the oscillator antenna partially extended and at a distance great enough to obtain a reading on the output meter. Make certain that the indicator lamp functions with the test oscillator adjusted for either 400, 1300, or 3000 cycles per second modulation.

NOTE: The antenna is tuned and sealed at the factory and is not to be adjusted in the field.

3.4 Services and Repair.- Servicing and repair details other than removal, substitution test, and replacement of tubes will not be permitted since they are functions of the Government repair depots and are not the responsibility of the aircraft manufacturer.

4. PREFLIGHT TESTS

4.1 General.- After the marker beacon receiving set has been installed in the aircraft, the following checks shall be made before placing it into service.

> NOTE: In some aircraft installations the marker beacon lamp is mounted on the pilot's instrument panel and is marked "MARKER BEACON"; however, in other installations the marker beacon lamp is a part of Indicator ID-48/ARN, Indicator ID-525/ARN or any of the following Course Indicators: ID-249/ARN, ID-249A/ARN, ID-249B/ARN, ID-351A/ARN or ID-387/ARN.

4.2 Test Equipment Required.- The following equipment shall be used for the preflight tests specified herein:

> Multimeter, AN/PSM-6, or equal
> Test Oscillator BC-376-( )

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4.3 Test Procedure.- The preflight tests shall be made as follows:

4.3.1 Preliminary Instructions.- Determine that the radio receiver is properly secured to Mounting MT-1546/ARN-32 and safety wired, and that the cable connector and radio frequency cable are securely fastened.

4.3.2 Power Supply.- The auxiliary supply shall be connected to the aircraft through the connection provided for that purpose.

4.3.3 Electrical Test.- The following test procedure shall be followed:

a. The marker beacon receiver shall be turned on for a warmup period of 5 minutes before any tests are made.

b. Adjust the direct-current supply voltage to 28 volts as measured at the aircraft bus. This voltage adjustment shall be made on the auxiliary power supply.

c. Place Test Oscillator BC-376-( ) on the ground approximately 15 feet from Antenna AT-536/ARN. Extend the test oscillator antenna to its full length and place parallel to the fore and aft axis of the aircraft.

d. Turn the test oscillator ON and adjust according to instructions on the inside of the cover plate. Set the modulation switch to 400.

e. Check to see that the indicator lamp is lit. Connect the headset to the pilot's headset jack. Set the pilot's interphone controls, if installed, for the reception of MARKER signals and ascertain that a 400 cycle signal is heard.

f. If for any reason the lamp fails to light, or no signals are heard in the pilot's headset, the aircraft antenna and lead-in may be bypassed by placing Test Oscillator BC-376-L in the aircraft. Using Cord CG-409/U, connect the rf output on the BC-376-L directly to the antenna connector on Receiver, Radio R-666 ( )/ARN-32 and repeat the procedure outlined in 4.3.3 d and e.

NOTE This can be made only when using Test Oscillator BC-376-L, as the earlier models were not equipped with a calibrated output attenuator.

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5 Flight Tests

5.1 General.- Unless otherwise directed by the procuring activity, flight tests shall be required on prototype installations only. Where flight tests are required, the actual operation of the marker beacon receiving set shall be checked as the aircraft is flown at various altitudes over one or more fan-type markers of a Civil Aeronautics Authority instrument landing system. The indicator lamp shall light or flash regularly depending on the type of transmission being received.

6 Notes

6.1 Intended Use.- The marker beacon receiving set is intended to give visual and aural indication to the pilot when his circraft passes over a marker beacon transmitter. The marker beacon receiving set operates at a crystal-controlled frequency of 75 megacycles. The marker beacon receiving set requires a primary power source of 28 volts direct current.

6.2 Definitions

6.2.1 Substitution Test.- A substitution test is defined as a test consisting of substituting a doubtful part or assembly in a satisfactorily operating equipment or major component for the purpose of checking the doubtful part for proper operation.

6.3 Handbooks.- It is necessary that personnel making tests described herein be thoroughly familiar with the operating characteristics of Receiving Set, Radio AN/ARN-32. The Handbook of Maintenance Instructions for Receiving Set, Radio AN/ARN-32 may be of interest in connection with this work.

NOTICE. When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner liscensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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